Siddharth Roy

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Education

National Institute of Technology, Jamshedpur August 2023 – Present
B.Tech in Electronics And Communication Engineering *Jamshedpur, Jharkhand*

Indian High School, Dubai April 2022 – May 2023
12th Grade, 75.6% *Dubai, U.A.E.*

Birla Public School, Pilani April 2020 – May 2021
10th Grade, 88.4% *Pilani, Rajasthan*

Relevant Coursework

Digital Electronics, Analog Electronics, Microprocessors and Microcontrollers, VLSI Design, AI/ML, Embedded Systems and IoT, Digital Communication, Analog Communication

Technical Skills

ML/AI: Machine Learning, OpenCV, PyTorch, TensorFlow
Languages & Frameworks: C/C++, Python, **ROS/ROS 2**, VHDL, Verilog, SystemVerilog
Robotics Concepts & Simulation: Data Structures & Algorithms, SLAM, Gazebo
Embedded Systems & RTOS: FreeRTOS, Embedded C, ARM Architecture
Hardware & Tools: Raspberry Pi, Arduino, ESP32, STM32, KiCad, Cadence, Vivado
Development Environment: Git, Linux, Shell Scripting, JTAG/SWD Debugging, Agile/Scrum

Experience

ROBOMANTHAN May. 2025 – July. 2025
Robotics and ML Intern *Link*

- **Built advanced IoT/IIoT solutions** using ESP32, Arduino (Uno, Mega, Nano), and sensor-integrated embedded systems; contributed to autonomous platforms including drones, UAVs, and rovers with ROS, Gazebo, and RViz for simulation and visualization.
- **Created intelligent robotic systems** by developing face recognition applications, programming humanoid robots, and enhancing hardware design skills with PCB development, Raspberry Pi programming, and hardware communication protocols.

Projects

VEGA (Capstone Project)
Vehicle Embedded Generalized Architecture

- Architected a modular automotive embedded platform using STM32F407 and ESP32 for real-time BLDC motor control, multi-sensor fusion, and CAN bus networking.
- Implemented PID control, FreeRTOS task scheduling, and custom CAN communication protocols for reliable, low-latency operation.
- Integrated hardware with PCB design, power management, and safety features (TVS, terminators, ADC filtering) ensuring robust, scalable architecture for EV and robotics applications.

MAVERICK (Team Horizons)
Modular Aerospace Embedded Flight Controller

- Prototyped a full-scale flight control system for rockets and UAVs using STM32H7 + ESP32-S3, integrating redundant IMUs, high-g accelerometer, barometer, GNSS, and pitot sensors for precision navigation.
- Developed PWM servo control, dual pyro ignition, LoRa + Wi-Fi/BLE telemetry, microSD logging, and CAN bus networking for autonomous and remote flight operations.
- Constructed PCB, power, and safety-critical systems with opto-isolated pyro drivers and modular architecture, enabling robust, scalable, and autonomous aerospace applications.

Achievements

- **E-Summit (E-Cell):** Won the Chat JPG event, a competitive Prompt Engineering-based challenge.
- **Ojass 2025:** Achieved 2nd place in the 7kg Robo War, demonstrating combat robotics skills.

Positions of Responsibility

- **Vice President** – Society of Electronics and Communication Engineering, NIT Jamshedpur.
- **Head of Electronics and Avionics** – Cosmology And Rocketry Club, NIT Jamshedpur.